untitled

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from collections import deque

# Part 1: simple reflex agent
def reflex_vacuum_agent(location, status):
    if status == "Dirty":
        return "Suck"
    return "Right" if location == "A" else "Left"

def run_reflex_agent(env, location, steps=6):
    print(f"Initial: room A = {env['A']}, room B = {env['B']}, vacuum in {location}")
    for t in range(1, steps + 1):
        action = reflex_vacuum_agent(location, env[location])
        if action == "Suck":
            env[location] = "Clean"
        elif action == "Right":
            location = "B"
        else:
            location = "A"
        print(f"t={t}: action = {action:<5} -> A = {env['A']:<5}, B = {env['B']:<5}, vacuum in {location}")

# Part 2: goal-based planner using BFS
def plan(location, dirt_a, dirt_b):
    start = (location, dirt_a, dirt_b)
    parent = {start: None}
    queue = deque([start])
    while queue:
        state = queue.popleft()
        loc, a, b = state
        if not a and not b:
            actions = []
            while parent[state] is not None:
                state, act = parent[state]
                actions.append(act)
            return actions[::-1]
        moves = {
            "Suck": (loc, a and loc != "A", b and loc != "B"),
            "Left": ("A", a, b),
            "Right": ("B", a, b),
        }
        for act, nxt in moves.items():
            if nxt not in parent:
                parent[nxt] = (state, act)
                queue.append(nxt)
    return None

if __name__ == "__main__":
    print("=== Simple reflex agent ===")
```

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========== RESTART
=== Simple reflex agent ===
Initial: room A = Dirty, room B = Dirty, vacuum in A
t=1: action = Suck  -> A = Clean, B = Dirty, vacuum in A
t=2: action = Right -> A = Clean, B = Dirty, vacuum in B
t=3: action = Suck  -> A = Clean, B = Clean, vacuum in B
t=4: action = Left  -> A = Clean, B = Clean, vacuum in A
t=5: action = Right -> A = Clean, B = Clean, vacuum in B
t=6: action = Left  -> A = Clean, B = Clean, vacuum in A

=== Goal-based planner (BFS) ===
Start: vacuum in A, A dirty = True , B dirty = True  -> ['Suck', 'Right', 'Suck']
Start: vacuum in A, A dirty = True , B dirty = False -> ['Suck']
Start: vacuum in A, A dirty = False, B dirty = True  -> ['Right', 'Suck']
Start: vacuum in A, A dirty = False, B dirty = False -> []
Start: vacuum in B, A dirty = True , B dirty = True  -> ['Suck', 'Left', 'Suck']
Start: vacuum in B, A dirty = True , B dirty = False -> ['Left', 'Suck']
Start: vacuum in B, A dirty = False, B dirty = True  -> ['Suck']
Start: vacuum in B, A dirty = False, B dirty = False -> []
>>
```